Squared Codegrees of Even Four-Uniform Hypergraphs

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Abstract

A four-uniform hypergraph is even if every set of five vertices spans an even number of edges. We determine the maximum asymptotic squared triple-codegree density of even hypergraphs to be 31/64. The lower bound is given by the classical bipartite matrix construction. The upper bound follows from an exact rational flag algebra certificate on seven vertices, with independently verified enumeration and coefficients. We also prove that the induced seven-vertex distribution of every near-extremal even hypergraph is concentrated on the ten types occurring in that construction. Finally, an elementary parity repair gives a quantitative obstruction to exceeding 31/64: an excess of η requires odd five-sets of density at least $\eta/2-o(1)$. The upper bound applies to all even hypergraphs, while the corresponding conjecture for arbitrary $K_5^{(4)}$-free hypergraphs remains open.

1 Introduction

For a four-uniform hypergraph, or 4-graph, H, let $e(H)$ be its number of edges. For a triple $T \subseteq V(H)$, its codegree $d_H(T)$ is the number of vertices $v \notin T$ for which $T \cup \{v\} \in E(H)$. We study the normalized squared-codegree sum

$$\sigma_n(H) = \frac{\sum_{T \in \binom{V(H)}{3}} d_H(T)^2}{\binom{n}{3}(n-3)^2}, \qquad n = |V(H)| \geq 4. \tag{1}$$

This parameter measures both edge density and variation among triple codegrees. Balogh, Clemen, and Lidický [1] study the corresponding extremal problem for forbidden hypergraphs. In particular, their Conjecture 8.2 asserts that the maximum asymptotic value of (1) over $K_5^{(4)}$-free hypergraphs is 31/64, where $K_5^{(4)}$ is the complete 4-graph on five vertices. They obtain this lower bound from the classical matrix construction and give the upper bound 0.55241250 for the unrestricted forbidden-clique problem.

A 4-graph is *even* if every five-vertex set spans an even number of edges. Thus an even 4-graph has zero, two, or four edges on every five-set and is automatically $K_5^{(4)}$-free. The matrix construction lies in this class. In simplicial terminology, even 4-graphs are 3-cocycles over the field with two elements. Kalai [4, Section 7.1] discusses extremal edge counts for these cocycles and reports Peled's flag algebra upper bound 0.6916 for their ordinary edge density. The objective in (1) is different: it counts two edges sharing a triple, up to a lower-order contribution from single edges.

Our main result establishes the conjectured squared-codegree value throughout the even class.

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Theorem 1.1. *For every even 4-graph H on $n \geq 7$ vertices,*

$$\sigma_n(H) \leq \frac{31}{64} + \frac{88}{n-5}.$$

Moreover,

$$\lim_{n\to\infty} \max\{\sigma_n(H) : |V(H)| = n,\ H \text{ is even}\} = \frac{31}{64}.$$

The finite error term is chosen for a transparent proof and is not optimized. The lower-bound construction is due to earlier work: Balogh, Clemen, and Lidický [1] attribute it to Giraud [3], while Kalai [4] describes it in connection with the earlier work of de Caen, Kreher, and Wiseman [2]. We reproduce its definition and squared-codegree calculation. The contribution of Theorem 1.1 is the matching upper bound for every even 4-graph, with no assumption on a partition, a matrix representation, or independence of its defining data.

The proof uses Razborov's flag algebra method [5]. Six positive semidefinite matrices yield an exact inequality on the 423 isomorphism classes of even 4-graphs on seven vertices. We specify the sampling of every matrix and give a finite averaging argument. The rational matrices, exhaustive enumeration, and independent verification programs are supplied in the accompanying `supplement` directory. No numerical optimization is needed to verify the final certificate.

The positive slack in this certificate also gives a local consequence. Let $\mathcal{G}$ be the family of seven-vertex 4-graphs obtainable by the matrix construction, with arbitrary bipartition and arbitrary binary matrix. This family consists of ten isomorphism types, described explicitly in Section 3. Write

$$\rho(H) = \frac{\left|\{U \in \binom{V(H)}{7} : H[U] \not\cong G \text{ for every } G \in \mathcal{G}\}\right|}{\binom{n}{7}}, \qquad \gamma = \frac{1}{6000},$$

where $H[U]$ denotes the induced hypergraph on U.

Theorem 1.2. *Every even 4-graph H on $n \geq 7$ vertices satisfies*

$$\sigma_n(H) + \gamma\rho(H) \leq \frac{31}{64} + \frac{88}{n-5}.$$

In particular, if H_n is even and $\sigma_n(H_n) \to 31/64$, then $\rho(H_n) \to 0$.

This is concentration of an induced distribution on seven vertices. It does not assert that every near-extremal hypergraph is close in edit distance to a matrix construction, or that its distribution converges to one fixed probability vector on the ten types.

For an arbitrary 4-graph H on $n \geq 5$ vertices, define its *parity defect* by

$$\varepsilon(H) = \frac{\left|\{U \in \binom{V(H)}{5} : e(H[U]) \text{ is odd}\}\right|}{\binom{n}{5}}.$$

A direct repair of parity turns Theorem 1.1 into a bound for all 4-graphs.

Theorem 1.3. *For every 4-graph H on $n \geq 7$ vertices,*

$$\sigma_n(H) \leq \frac{31}{64} + 2\varepsilon(H) + \frac{88}{n-5}.$$

Consequently, if $\eta > 0$ and $\liminf_{n\to\infty} \sigma_n(H_n) \geq 31/64 + \eta$, then $\liminf_{n\to\infty} \varepsilon(H_n) \geq \eta/2$.

For $K_5^{(4)}$-free hypergraphs, the odd five-sets in this statement have one or three edges. Thus a counterexample to the conjecture of Balogh, Clemen, and Lidický with a fixed positive excess must contain a positive density of those configurations. The present result does not control their contribution sharply enough to settle that conjecture.

Section 2 proves the parity representation, repair estimate, and known lower bound. Section 3 describes and verifies the finite certificate. Section 4 transfers it to arbitrary even hypergraphs and proves the three theorems.

2 Parity and the matrix construction

We first record an elementary representation of even hypergraphs. It explains the scope of the evenness assumption and will later allow us to compare an arbitrary hypergraph with an even one. For a function $g : \binom{V}{3} \to \{0,1\}$, define a 4-graph δg by

$$\mathbf{1}_{E(\delta g)}(X) = \sum_{T \in \binom{X}{3}} g(T) \pmod 2 \qquad (X \in \tbinom{V}{4}).$$

Here g is a 2-cochain and δg is its coboundary.

Lemma 2.1. *A 4-graph is even if and only if it equals δg for some function g on triples. Furthermore, every 4-graph H on $n \geq 5$ vertices has an even 4-graph H' on the same vertex set such that*

$$\frac{|E(H) \triangle E(H')|}{\binom{n}{4}} \leq \varepsilon(H). \tag{2}$$

Proof. In the sum of the edge indicators of δg over a five-set, every triple occurs twice. The sum is therefore zero modulo two, so δg is even.

Conversely, fix a vertex v of an even 4-graph H. Set $g(T) = 0$ when $v \in T$, and set $g(T) = \mathbf{1}_{E(H)}(T \cup \{v\})$ otherwise. On a four-set containing v, exactly one summand in the definition of δg can be nonzero, and it equals the edge indicator of H. On a four-set X avoiding v, the parity equation on $X \cup \{v\}$ gives

$$\mathbf{1}_{E(H)}(X) = \sum_{T \in \binom{X}{3}} \mathbf{1}_{E(H)}(T \cup \{v\}) \pmod 2.$$

Hence $H = \delta g$.

For a general H, use the same definition of g and put $H' = \delta g$. The two hypergraphs agree on four-sets containing v. On a four-set X avoiding v, they disagree precisely when $X \cup \{v\}$ spans an odd number of edges in H. The average number of such odd five-sets containing v is

$$\frac{5}{n}\varepsilon(H)\binom{n}{5} = \varepsilon(H)\binom{n-1}{4}.$$

Choose v whose count is at most this average. The resulting H' is even and has at most $\varepsilon(H)\binom{n-1}{4}$ disagreements with H, which implies (2). □

We next recall the matrix construction. Partition V into $A \sqcup B$, and assign a bit $m_{ab} \in \{0,1\}$ to each pair $(a,b) \in A \times B$. Include every four-set meeting A in one or three vertices. For distinct $a, a' \in A$ and $b, b' \in B$, include $\{a, a', b, b'\}$ if

$$m_{ab} + m_{ab'} + m_{a'b} + m_{a'b'} = 1 \pmod 2. \tag{3}$$

No four-set contained in one part is an edge. Denote the resulting hypergraph by $M(A, B; m)$.

Proposition 2.2. *Every $M(A,B;m)$ is even. If $|A| = \lfloor n/2 \rfloor$ and $|B| = \lceil n/2 \rceil$, then there are choices of m for which*

$$\sigma_n(M(A,B;m)) \geq \frac{31}{64} - O(n^{-1}).$$

Proof. A five-set contained in one part spans no edges. A five-set meeting one part in four vertices spans four edges. It remains to consider a five-set with three vertices in one part and two in the other. Two of its four-subsets meet the first part in three vertices and are edges. The other three edge indicators are the three rectangular parities from (3). Their sum is zero modulo two, since each cross-pair bit occurs twice. Thus zero or two of them are edges, and the total is even.

Choose the bits independently and uniformly. Write $a = |A|$ and $b = |B|$. A triple contained in A has codegree b, and a triple contained in B has codegree a. For a triple containing two vertices of A and one of B, the codegree has the distribution

$$a - 2 + Z, \qquad Z \sim \mathrm{Bin}(b-1, 1/2).$$

Indeed, the $a-2$ remaining vertices of A always extend it to an edge. Conditional on the two matrix entries already incident with its vertex in B, the rectangular parities for the other vertices of B are independent fair bits. Therefore

$$\mathbb{E}(a-2+Z)^2 = \left(a - 2 + \frac{b-1}{2}\right)^2 + \frac{b-1}{4}.$$

Interchanging a and b gives the corresponding formula for a triple containing one vertex of A and two of B. Summing over the four kinds of triples and dividing by the denominator in (1), we obtain

$$\mathbb{E}\sigma_n(M(A,B;m)) = \frac{1}{4}\left(\frac{1}{2}\right)^2 + \frac{3}{4}\left(\frac{3}{4}\right)^2 + O(n^{-1}) = \frac{31}{64} + O(n^{-1}).$$

Some choice of the bits has value at least this expectation. □

Finally, changing a small fraction of edges changes the objective by at most twice that fraction.

Lemma 2.3. *For 4-graphs H, H' on the same $n \geq 4$ vertices,*

$$|\sigma_n(H) - \sigma_n(H')| \leq 2\frac{|E(H) \triangle E(H')|}{\binom{n}{4}}.$$

Proof. Toggling one edge changes exactly four triple codegrees by one. Each lies between zero and $n-3$, so the change in its square is at most $2(n-3)$. Toggling the differing edges one at a time gives a total change in the numerator of (1) of at most $8(n-3)|E(H) \triangle E(H')|$. Now use $\binom{n}{3}(n-3) = 4\binom{n}{4}$. □

3 An exact certificate on seven vertices

The upper bound is a finite inequality averaged over seven-vertex induced subgraphs. We describe the host enumeration, the six matrices of joint flag probabilities, and the exact arithmetic certificate. This description also fixes all normalizations needed in Section 4.

For an n-vertex 4-graph with $n \geq 5$, put

$$q(H) = \frac{\sum_{T \in \binom{V(H)}{3}} d_H(T)(d_H(T)-1)}{\binom{n}{3}(n-3)(n-4)}, \qquad D(H) = \frac{e(H)}{\binom{n}{4}}.$$

Thus $q(H)$ is the probability that a uniform triple and two ordered distinct extension vertices form two edges. Since $\sum_T d_H(T) = 4e(H)$,

$$\sigma_n(H) = \frac{n-4}{n-3} q(H) + \frac{D(H)}{n-3} \leq q(H) + \frac{1}{n-3}. \tag{4}$$

If U is a uniform seven-subset of $V(H)$, then

$$\mathbb{E} q(H[U]) = q(H). \tag{5}$$

Both sides sample the same configuration on five distinct vertices; choosing two additional vertices does not change its distribution.

For the finite enumeration we label vertices $0, 1, \ldots, 6$. Represent a 4-graph F by the complementary 3-graph $\overline{F}^*$, whose triples are the complements of the edges of F. Evenness of F is equivalent to every pair of $\overline{F}^*$ having even codegree: the edges of F inside the five-set complementary to a pair correspond exactly to the triples containing that pair.

The 20 triples contained in $\{0, \ldots, 5\}$ can be selected freely. The indicator of $\{i, j, 6\}$ is then forced to equal, modulo two, the number of selected triples containing $\{i, j\}$. This makes the codegree of every pair avoiding 6 even. For a pair $\{i, 6\}$, the sum of the newly assigned indicators is also even, because each originally selected triple through i is counted twice. Hence this procedure produces exactly 2^{20} labelled even hosts. Their permutation orbits comprise 423 isomorphism classes.

The data file orders the complementary triples by listing first the 20 triples on $\{0, \ldots, 5\}$ in lexicographic order and then the 15 triples $\{i, j, 6\}$, again lexicographically in (i, j). If b_j is the indicator of the jth triple, starting at $j = 0$, its mask is $\sum_{j=0}^{34} 2^j b_j$. A representative is specified by this integer. For such a host,

$$q(F) = \frac{1}{420} \sum_{P \in \binom{\{0,\ldots,6\}}{2}} d_{\overline{F}^*}(P)(d_{\overline{F}^*}(P) - 1).$$

To see the denominator, a five-set containing j edges contributes $j(j-1)$ ordered pairs of edges sharing a triple, and there are $\binom{7}{5} \cdot 20 = 420$ possible ordered choices on five vertices.

For each host F, define six symmetric matrices $M_b(F)$, $0 \leq b \leq 5$. In every block, average over all 7! ordered permutations $r_0, \ldots, r_6$ of its vertices. A sampled pair of flags contributes one to the corresponding matrix entry; division by $7! = 5040$ gives $M_b(F)$. A flag records the induced edges on a labelled root and an extension. The following rules determine its index completely.

Three roots. The root is (r_0, r_1, r_2), and the two extensions are $\{r_3, r_4\}$ and $\{r_5, r_6\}$. On each resulting five-set, encode the four-edges by a five-bit mask according to the omitted vertex. The first three positions are the labelled roots; the last two positions can be exchanged. The mask has even parity. Replace it by the smaller of the two integers obtained by exchanging the last two bits. There are 12 possible canonical masks, ordered increasingly. This defines the 12×12 matrix $M_0(F)$.

Four roots. The root is (r_0, r_1, r_2, r_3), with edge indicator $t \in \{0, 1\}$, and the extensions are r_4, r_5; the vertex r_6 is unused. For an extension v, its four-bit mask has bit i equal to the edge indicator of $\{r_0, r_1, r_2, r_3\} \setminus \{r_i\} \cup \{v\}$. The mask has parity t. The eight masks of each parity are ordered increasingly. The permutation contributes to $M_1(F)$ if $t = 0$ and to $M_2(F)$ if $t = 1$, and contributes zero to the other block. Both matrices have order 8.

Five roots. The root is $(r_0, \ldots, r_4)$ and the extensions are r_5, r_6. Encode its five four-edge indicators by the omitted root vertex and call the resulting mask u. There are three blocks, using precisely

the labelled masks $u = 0, 3, 15$. A permutation contributes to $M_3(F), M_4(F), M_5(F)$ respectively when its root mask equals that specified integer; if its root mask is different, it contributes zero to that block.

For an extension v, its ten-bit mask, indexed by the lexicographically ordered pairs $\{i, j\} \subseteq \{0, \dots, 4\}$, records whether $\{r_0, \dots, r_4\} \setminus \{r_i, r_j\} \cup \{v\}$ is an edge. Regard this mask as a graph on the five root positions. The parity of the degree at position i equals bit i of u. Indeed, in the complementary graph of the induced six-vertex 4-graph all degrees are even, and the edge from root i to v has indicator u_i. For each specified u, exactly $2^{10-4} = 64$ masks have the required degree parities. Order them increasingly. These define three matrices of order 64. Their normalization is still 5040: we do not condition on the root having the specified mask.

Exchanging the two extensions preserves the sampling in each block, so all six matrices are symmetric. For matrices of the same size, write $\langle A, B\rangle = \sum_{i,j} A_{ij}B_{ij}$.

Lemma 3.1 (Exact certificate)**.** *There are rational positive semidefinite matrices $Q_0, \dots, Q_5$ of orders $12, 8, 8, 64, 64, 64$, respectively, and rational numbers $s(F) \geq 0$ for every seven-vertex even host, such that*

$$\frac{31}{64} - q(F) = \sum_{b=0}^{5} \langle Q_b, M_b(F)\rangle + s(F). \tag{6}$$

The slack is zero exactly for $F \in \mathcal{G}$, and otherwise $s(F) \geq \gamma$. If $B_b = \max_i (Q_b)_{ii}$, then

$$1 + 8B_0 + 2\sum_{b=1}^{5} B_b < 88. \tag{7}$$

Proof. This is a finite, computer-assisted lemma. We give the precise verification procedure so that the conclusion does not depend on the optimization used to find the matrices. The accompanying file `even7_exact_certificate.json` specifies each matrix in the form

$$Q_b = K_b S_b K_b^{\mathsf{T}},$$

where K_b is integral and S_b is rational and symmetric. The orders of S_b are $8, 4, 3, 63, 63, 57$. Clear its denominators by a positive integer to obtain an integer symmetric matrix A. Fraction-free elimination starts with previous pivot $p = 1$ and, at stage k, replaces the trailing entries by

$$A_{ij} \longleftarrow \frac{A_{kk}A_{ij} - A_{ik}A_{kj}}{p} \quad (i, j > k), \qquad p \longleftarrow A_{kk}.$$

All divisions are exact. Every successive pivot is positive, and these pivots are the leading principal minors of the original integer matrix. Sylvester's criterion gives $S_b \succ 0$, hence $Q_b \succeq 0$.

For every one of the 423 representatives, the verifier reconstructs Q_b over the rationals and checks (6) with the stored integer flag counts divided by 5040. Every resulting slack is nonnegative. There are ten zero slacks, and every positive slack is greater than $\gamma = 1/6000$, as checked by exact rational comparison. The largest diagonal entries are, in block order,

$$\left(\frac{8896637}{31250000}, \frac{105531}{10^9}, \frac{113931}{10^9}, \frac{3930324933}{10^9}, \frac{21878641233}{10^9}, \frac{16485478529}{10^9}\right),$$

which give (7) by rational arithmetic.

The coefficient data are checked separately. The program in `independent_audit` represents hosts directly by four-edges. For each supplied representative, it examines every permutation,

Table 1: The ten matrix-construction types on seven vertices. The mask uses the complementary triple order specified above; $420q(F)$ is an integer. Probabilities refer to independent fair part assignments and cross-pair bits.

Complementary mask	Orbit size	$420q(F)$	Probability
0	1	0	1/64
34326184959	21	140	21/1024
6156266679	7	180	7/64
28201529160	35	72	35/4096
28839063387	420	162	105/1024
30013468542	315	192	315/4096
34326261759	105	200	105/1024
16641128943	210	230	105/512
24694192863	1260	222	315/1024
33671344093	30	252	105/2048

verifies evenness and its orbit size, and records its edges containing a fixed vertex. These records cover all 2^{20} possible anchored edge assignments, with no overlap between different representatives. Thus the list is exhaustive and has no repeated isomorphism class. The program then reconstructs every flag count from the sampling rules above and recomputes the objective from triple codegrees. It checks all $423(12^2 + 2 \cdot 8^2 + 3 \cdot 64^2) = 5{,}312{,}880$ integer entries against the stored data. A second exact-arithmetic checker evaluates the certificate on these independently reconstructed entries.

It remains to identify the zero-slack hosts with $\mathcal{G}$. Table 1 lists their complementary masks. To enumerate the matrix construction on seven vertices, choose a part size $a \in \{0, \dots, 7\}$, put the first a vertices in that part, and examine each of the $2^{a(7-a)}$ binary matrices. Classify each resulting host by its permutation orbit. This enumeration produces exactly the ten listed types. The program `verify_matrix_support.py` performs this check independently from the flag calculations. It also verifies the probabilities in the table by weighting each matrix of part size a by $2^{-7-a(7-a)}\binom{7}{a}$. This is the distribution obtained by independent fair part assignments and independent fair cross-pair bits. All ten probabilities are positive, proving the asserted equivalence. □

The supplied verification uses only exact integer and rational arithmetic. The basic checker needs Python 3 and its standard library. Independent reconstruction additionally needs a C++17 compiler. The supplement contains all input data and has no dependency on a semidefinite programming solver or on a separate research directory.

4 Averaging and consequences

The matrices $M_b(F)$ record disjoint extensions. Positive semidefiniteness is immediate for independent extensions, so the final step is to bound the difference between these two sampling schemes. We retain the slack throughout this argument to prove the local conclusion together with the main upper bound.

Lemma 4.1. *Let H be even on $n \geq 7$ vertices, and let U be a uniform seven-subset. For the matrices in Lemma 3.1,*

$$\sum_{b=0}^{5} \mathbb{E}\langle Q_b, M_b(H[U])\rangle \geq -\frac{8B_0}{n-3} - \frac{2(B_1+B_2)}{n-4} - \frac{2(B_3+B_4+B_5)}{n-5}.$$

Proof. Fix an ordered root of the appropriate size. For the three-root block, choose an unordered pair uniformly from its $n-3$ remaining vertices. Let p be the vector of probabilities of its 12 flag types. If two such pairs are chosen independently, the expected joint matrix is pp^{T}, and

$$\langle Q_0, pp^{\mathsf{T}} \rangle = p^{\mathsf{T}} Q_0 p \geq 0.$$

For a four-root or five-root block, use independent uniform single-vertex extensions. Define p to be zero when the root does not have that block's prescribed type. The same identity gives nonnegativity after averaging over all ordered roots. In particular, the five-root blocks use unconditional root sampling, exactly as in their definition.

For any positive semidefinite Q_b, its two-by-two principal minors imply

$$|(Q_b)_{ij}| \leq \sqrt{(Q_b)_{ii}(Q_b)_{jj}} \leq B_b.$$

The scalar contribution of a sampled pair of flags is therefore between $-B_b$ and B_b. Couple the independent choices with choices conditioned to be disjoint by retaining them when they are disjoint and otherwise resampling from the conditioned distribution. If the independent choices intersect with probability c, the two expected scalar contributions differ by at most $2B_b c$.

For two independent pairs among $n-3$ vertices, the expected intersection size is $4/(n-3)$, so $c \leq 4/(n-3)$. For independent singletons extending four or five roots, the collision probabilities are $1/(n-4)$ and $1/(n-5)$, respectively. Conditioned on disjointness, the root and extensions have exactly the distribution obtained by first choosing a uniform seven-set and then a uniform permutation. The unused vertex in the four-root blocks does not change their marginal distribution. Applying the three error bounds to the six blocks and summing proves the assertion. □

Proof of Theorems 1.1 and 1.2. Average (6) over the induced seven-vertex subgraphs of H. Equations (5) and the slack bound give

$$\frac{31}{64} - q(H) = \sum_{b=0}^{5} \mathbb{E}\langle Q_b, M_b(H[U]) \rangle + \mathbb{E}s(H[U]), \qquad \mathbb{E}s(H[U]) \geq \gamma\rho(H).$$

By Lemma 4.1,

$$q(H) + \gamma\rho(H) \leq \frac{31}{64} + \frac{8B_0}{n-3} + \frac{2(B_1+B_2)}{n-4} + \frac{2(B_3+B_4+B_5)}{n-5}.$$

Now (4) and (7) imply

$$\sigma_n(H) + \gamma\rho(H) \leq \frac{31}{64} + \frac{1 + 8B_0 + 2\sum_{b=1}^{5} B_b}{n-5} < \frac{31}{64} + \frac{88}{n-5}.$$

This proves Theorem 1.2 and the upper bound in Theorem 1.1. Proposition 2.2 gives a sequence of even hypergraphs attaining 31/64 asymptotically, so the stated extremal limit exists and has that value. Since $\gamma > 0$, the distributional conclusion follows from the same inequality. □

Proof of Theorem 1.3. Lemma 2.1 gives an even H' differing from H on at most an $\varepsilon(H)$ fraction of four-sets. By Lemma 2.3 and Theorem 1.1,

$$\sigma_n(H) \leq \sigma_n(H') + 2\varepsilon(H) \leq \frac{31}{64} + 2\varepsilon(H) + \frac{88}{n-5}.$$

Rearranging and taking lower limits gives the final assertion. □

The local statement identifies the possible seven-vertex support of an extremal sequence, but does not reconstruct a global bipartition or cross-pair matrix. Such a reconstruction would require an additional consistency argument between overlapping vertex sets. A separate obstruction remains for the unrestricted $K_5^{(4)}$ problem: odd five-sets are allowed, and Theorem 1.3 only provides a necessary condition on their density. These are distinct questions left by the present result.

Declaration of AI use

The author used GPT-6.1 to assist with exploring proof strategies, checking calculations, and improving the exposition. The author is responsible for the mathematical arguments, computational certificates, and conclusions presented in this paper.

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